THE DENSITY OF ADMISSIBLE n IN ERDŐS–SELFRIDGE PROBLEM 1063, WITH A GROWTH CONJECTURE FOR n_k

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ABSTRACT. For $k \geq 2$ let $n_k \geq 2k$ be the least n such that $n-i$ divides $\binom{n}{k}$ for all but exactly one $0 \leq i < k$ (Erdős problem 1063, OEIS A389360). We describe, prime by prime, which index fails: $(n-i) \nmid \binom{n}{k}$ iff $v_p(n-i) > M_p(n) - d_p(n)$ for some prime $p \leq k$, where M_p is the largest p-adic valuation in the block $\{n-k+1, \ldots, n\}$ and d_p counts the non-carries in the lowest $\lfloor \log_p k \rfloor$ digits when k and $n-k$ are added in base p. This gives a two-line proof of the Erdős–Selfridge observation and the explicit lower bound $n_k - i_0 \geq \prod_{p^a \| k} p^{a+\lfloor \log_p(k-1) \rfloor}$ for the failing index i_0 (with equality for $k = 4, 6, 14$), hence $n_k \geq \prod_{p^a \| k} p^{a+\lfloor \log_p(k-1) \rfloor}$; this bound is formally verified in Lean 4. The admissible n form a periodic set whose density $\delta(k)$ is an explicit finite product-sum, and we prove

$$\log \frac{1}{\delta(k)} = (C + o(1)) \frac{k}{\log k}, \qquad C = \sum_{j \geq 1} \frac{1}{j} \log\Big(1 + \frac{1}{j}\Big) = 1.25774688694\ldots$$

We conjecture that $n_k \delta(k)$ behaves like an exponential random variable, so that $\log n_k = (C + o(1))k/\log k$, and support this with the values n_k for $60 \leq k \leq 110$ (extending the OEIS table, which ends at $k = 59$, and the values for $k \leq 75$ posted in the discussion thread), two pre-registered out-of-sample tests, and a correlation of 0.988 between $\log n_k$ and $\log(1/\delta(k))$ for $31 \leq k \leq 110$. The conjecture lies strictly between the bounds $c(\log k)^2$ and $(1+o(1))\, k \log\log k/\log k$ recently claimed by Cipollini, and predicts that the upper one is off by a factor of order $\log\log k$.

1. INTRODUCTION

Erdős and Selfridge [6] observed that for $n \geq 2k$ at least one of the k consecutive integers $n, n-1, \ldots, n-k+1$ fails to divide $\binom{n}{k}$; a proof was given by Monier [10]. Problem 1063 on `erdosproblems.com` [5] (also B31 in Guy's collection [8]) asks to estimate

$$n_k = \min\{\, n \geq 2k : \ (n-i) \mid \tbinom{n}{k} \text{ for all but exactly one } 0 \leq i < k \,\}.$$

Monier showed $n_k \leq k!$ for $k \geq 3$, and Cambie improved this to $n_k \leq k \cdot \operatorname{lcm}(2, \ldots, k-1) = e^{(1+o(1))k}$. The OEIS entry A389360 [12] lists n_k for $k \leq 59$. In the problem's discussion thread [5], the user "rickyc" proved $n_k \geq \max\big(2k, \prod_{p^a \| k} p^{a+\lfloor \log_p(k/2) \rfloor}\big)$ (26 June 2026, with GPT-5.5 Thinking), and then the $k-1$ version in the sharper form $p^{\,a+v_p\binom{k-1}{e}+\lfloor \log_p \max(e,k-1-e) \rfloor} \mid n-e$ for the failing index e (10 July 2026, with GPT-5.6 Sol). Our Proposition 3.1 is an independent proof of the $k-1$ bound; what is new here is the identification of the equality cases, the characterisation of the bad set behind it, and a formal verification in Lean 4. Two proof claims by Cipollini [1, 2], listed on the problem page (which notes that listing is no guarantee of correctness), assert

$$c(\log k)^2 \ \leq \ \log n_k \ \leq \ \frac{k}{\log k}\Big(\log\log k + \log\log\log k + \log 2 + o(1)\Big). \tag{1}$$

A bound of the same shape, $\log n_k \leq Ck \log\log k/\log k$, was announced independently two days earlier by White [13]; an upper bound of exactly the form $n_k \ll \exp\big(\frac{k}{\log k}(\log\log k + \log\log\log k + \log 2)\big)$ is recorded in DeepMind's formal-conjectures repository as formally proved in Lean 4 by

Date: October 22, 2026.

the LEAP agent [9, 7]; we have not re-checked that proof. No lower bound for n_k appears among the formalized results there.

Throughout, $k \geq 2$ and $n \geq 2k$ are integers, the *block* is $B = \{n-k+1, \ldots, n\}$, and the index $i \in [0, k)$ refers to the element $n - i$. We call i *bad* if $(n-i) \nmid \binom{n}{k}$, and n *admissible* if it has exactly one bad index. For a prime p and an integer $x \neq 0$, $v_p(x)$ is the p-adic valuation, and

$$L_p = L_p(k) = \lfloor \log_p k \rfloor, \qquad M_p(n) = \max_{0 \leq i < k} v_p(n-i),$$

$$d_p(n) = \#\{1 \leq e \leq L_p : \ n \bmod p^e \ \geq \ k \bmod p^e\}.$$

Note that $d_p(n)$ counts the levels $e \leq L_p$ at which adding k and $n-k$ in base p produces no carry.

Our results are the following.

Theorem 1.1 (local description)**.** *For every prime $p \leq k$, $v_p\binom{n}{k} = M_p(n) - d_p(n)$, and an index i is bad if and only if $v_p(n-i) > M_p(n) - d_p(n)$ for some prime $p \leq k$. If $p \mid k$ then $d_p(n) \geq v_p(k) \geq 1$, so every index of largest p-adic valuation is bad; in particular some index is always bad.*

Theorem 1.2 (explicit lower bound)**.** *If n is admissible with bad index i_0, then*

$$W'_k := \prod_{p^a \| k} p^{\,a + \lfloor \log_p(k-1) \rfloor} \quad \textit{divides} \quad n - i_0.$$

Hence $n_k \geq W'_k$. Equality $n_k - i_0 = W'_k$ holds for $k = 4, 6, 14$ and for no other $k \leq 110$.

Theorem 1.3 (exact density)**.** *Admissibility of n depends only on $n \bmod p^{2L_p}$ for the primes $p \leq k$. The admissible integers therefore form a union of residue classes modulo $D_k = \prod_{p \leq k} p^{2L_p}$, of density*

$$\delta(k) = \sum_{i_0=0}^{k-1} \prod_{p \leq k} \Pr\left(E_{p,i_0}\right), \qquad E_{p,i_0} = \{n : \ \mathrm{Bad}_p(n) \subseteq \{i_0\}\}, \tag{2}$$

where $\mathrm{Bad}_p(n) = \{i : \ v_p(n-i) > M_p(n) - d_p(n)\}$ and $\Pr(E_{p,i_0})$ is the proportion of residue classes modulo p^{2L_p} in E_{p,i_0}. For $p > \sqrt{k}$,

$$\Pr(E_{p,i_0}) = \frac{k \bmod p}{p} + \left[\, i_0 \bmod p \ \geq \ k \bmod p \,\right] \cdot \begin{cases} 1/p & \text{if } \lfloor k/p \rfloor = 1, \\ 1/p^2 & \text{if } \lfloor k/p \rfloor \geq 2. \end{cases} \tag{3}$$

Theorem 1.4 (size of the density)**.** *Let $S(k) = \sum_{\sqrt{k} < p \leq k,\ p \nmid k} \log \dfrac{p}{k \bmod p}$. Then*

$$\log \frac{1}{\delta(k)} = S(k) + O(\sqrt{k}) \qquad \textit{and} \qquad S(k) = (C + o(1)) \, \frac{k}{\log k},$$

where

$$C = \int_1^\infty \frac{-\log\{x\}}{x^2}\, dx = \sum_{j \geq 1} \frac{1}{j} \log\left(1 + \frac{1}{j}\right) = \sum_{m \geq 1} \frac{(-1)^{m+1} \zeta(m+1)}{m} = 1.25774688694436963\ldots$$

Consequently $\log(1/\delta(k)) = (C + o(1))\, k / \log k$.

The density is a statement about the set of admissible n, not about its least element. The following is a conjecture.

Conjecture 1.5. *$n_k\,\delta(k)$ is approximately an* Exp(1) *random variable, so that typically* $\log n_k = \log(1/\delta(k)) + O(1)$. *Under this heuristic, and only under it,*

$$\log n_k = (C + o(1))\,\frac{k}{\log k}.$$

Section 7 reports the evidence: the values n_k for $60 \le k \le 110$ (Table 2), two pre-registered out-of-sample tests in which 47 of 51 values fell inside the central 90% interval of $\mathrm{Exp}(1)/\delta(k)$ (46 expected), and the statistics of $\log n_k - \log(1/\delta(k))$. Section 8 explains why the construction behind the upper bound in (1) loses a factor of order $\log\log k$ relative to the conjecture.

Provenance. The analysis, the search code and the first draft of this paper were produced with Claude (Anthropic) at the author's direction. The mathematics was independently refereed by further AI reviewers, including Claude Fable 5.1, Gemini 4 Argon and a Meta AI model, using independently written code; every value in Table 2 was verified by exact integer arithmetic, and minimality was confirmed by two independently written exhaustive searches. Code, data and the pre-registered prediction files are available at [3].

2. The bad set, prime by prime

Lemma 2.1. *For $0 \le i < k$: $(n-i) \mid \binom{n}{k}$ iff $v_p(n-i) \le v_p\binom{n}{k}$ for every prime $p \le k$.*

Proof. With $P = \prod_{j<k}(n-j) = k!\binom{n}{k}$ we have $(n-i) \mid \binom{n}{k} \iff k! \mid P/(n-i)$. For $p > k$, $v_p(k!) = 0$, so there is no condition. For $p \le k$ the condition $v_p(P) - v_p(n-i) \ge v_p(k!)$ reads $v_p(n-i) \le v_p(P) - v_p(k!) = v_p\binom{n}{k}$. □

So the set of bad indices is $\bigcup_{p \le k} \mathrm{Bad}_p(n)$ with $\mathrm{Bad}_p(n) = \{i : v_p(n-i) > v_p\binom{n}{k}\}$.

Lemma 2.2. *For $q = p^e$, the block contains $\lfloor k/q \rfloor + \delta_e$ multiples of q, where $\delta_e = \delta_e(n) = [\,n \bmod q < k \bmod q\,]$. Hence $v_p\binom{n}{k} = \sum_{e \ge 1} \delta_e$.*

Proof. Write $n = aq + r$ and $k = bq + s$ with $0 \le r, s < q$. The number of multiples of q in the block is $\lfloor n/q \rfloor - \lfloor (n-k)/q \rfloor$, which equals b if $r \ge s$ and $b+1$ if $r < s$. Summing over e and using Legendre's formula, $v_p(P) = \sum_e (\lfloor k/p^e \rfloor + \delta_e) = v_p(k!) + \sum_e \delta_e$. □

Thus $d_p(n) = \#\{1 \le e \le L_p :\ \delta_e(n) = 0\}$.

Lemma 2.3. $v_p\binom{n}{k} = M_p(n) - d_p(n)$. *Consequently:*

1. $\mathrm{Bad}_p(n) = \varnothing$ *iff* $d_p(n) = 0$;
2. *if $d_p(n) \ge 1$, every index of maximal p-adic valuation is bad, and $|\mathrm{Bad}_p(n)| = 1$ iff the maximum is attained at a single index i^* and every other index has valuation $\le M_p - d_p$.*

Proof. Write $L = L_p$. The block has $k \ge p^L$ elements, so it contains a multiple of p^L and $M_p \ge L$. For $e > L$ we have $p^e > k$, so $\delta_e = [\text{the block contains a multiple of } p^e] = [M_p \ge e]$. Hence $\sum_e \delta_e = (L - d_p) + (M_p - L) = M_p - d_p$. Parts (1) and (2) follow from $\mathrm{Bad}_p(n) = \{i : v_p(n-i) > M_p - d_p\}$. □

Corollary 2.4 (Erdős–Selfridge)**.** *If $p \mid k$ then $d_p(n) \ge v_p(k) \ge 1$ for every n, because $k \bmod p^e = 0$ for $e \le v_p(k)$ (and $v_p(k) \le L_p$). So $\mathrm{Bad}_p(n) \ne \varnothing$, and for every $n \ge k$ some $n - i$ with $0 \le i < k$ fails to divide $\binom{n}{k}$.* □

This proves Theorem 1.1.

3. The explicit lower bound

Proposition 3.1. *Let n be admissible with bad index i_0, and let $p^a \| k$. Then $n - i_0$ is the unique element of the block with maximal p-adic valuation, and*

$$v_p(n - i_0) \;\geq\; a + \lfloor \log_p(k-1) \rfloor \;=\; a + L_p - [\, k = p^{L_p} \,].$$

Proof. Write $L = L_p$. By Corollary 2.4, $\mathrm{Bad}_p(n) \neq \varnothing$, so $\mathrm{Bad}_p(n) = \{i_0\}$, and by Lemma 2.3(2) i_0 is the unique index of maximal valuation and every other index has valuation at most $M_p - d_p$.

Let e_1 be the largest e with $0 \leq e \leq L$ such that the block contains at least two multiples of p^e ($e = 0$ always qualifies); one of them is not $n - i_0$, so the second-largest valuation in the block is at least e_1, whence $M_p - d_p \geq e_1$. For every $e \in (e_1, L]$ the block contains exactly one multiple of p^e; since $\lfloor k/p^e \rfloor \geq 1$, Lemma 2.2 forces $\lfloor k/p^e \rfloor = 1$ and $\delta_e = 0$.

Suppose $k \neq p^L$. Since $p^a \mid k$, the block contains exactly $k/p^a \geq 2$ multiples of p^a, so $e_1 \geq a$. The levels $e \leq a$ (where $k \bmod p^e = 0$, so $\delta_e = 0$) and the levels $e \in (e_1, L]$ are disjoint sets of non-carries, so $d_p \geq a + (L - e_1)$, and $M_p \geq e_1 + d_p \geq a + L$.

If $k = p^L$, then $a = L$, $d_p = L$, and the block contains $p \geq 2$ multiples of p^{L-1}, so the second-largest valuation is at least $L - 1$ and $M_p \geq (L-1) + L = 2L - 1 = a + \lfloor \log_p(k-1) \rfloor$. □

Proof of Theorem 1.2. By Proposition 3.1, $p^{a + \lfloor \log_p(k-1) \rfloor} \mid n - i_0$ for every $p^a \| k$, so $W'_k \mid n - i_0$ and $n_k \geq n_k - i_0 \geq W'_k$. The equality cases were checked against Table 2 and the OEIS values: $n_4 - 1 = 8$, $n_6 - 3 = 72$, $n_{14} - 9 = 784$, and $n_k - i_0 > W'_k$ for every other $k \leq 110$. □

Formal verification. Proposition 3.1 (in the slightly stronger form that uses only the divisibility of the elements $n - i$, $i \neq i_0$), the divisibility $W'_k \mid n - i_0$, and the unconditional bound $W'_k \leq n_k$ for $k \geq 3$ have been formally verified in Lean 4 with Mathlib, using the definition of n_k from [7] verbatim. The same file proves rickyc's sharper divisibility $p^{\,a + v_p\binom{k-1}{e} + \lfloor \log_p \max(e, k-1-e) \rfloor} \mid n - e$ for every $p^a \| k$, and that the failing index is $e = n \bmod k$; together these are the structural input of the search of jbell31, whose minimality claims therefore no longer rest on an unformalized step. It also proves Cambie's bound $n_k \leq k \,\mathrm{lcm}(1, \ldots, k-1)$ independently of [9]; it is part of [3]. The case $k = 2$ is immediate ($W'_2 = 2 < 4 = n_2$).

Remark 3.2. (i) Since $\lfloor \log_p(k-1) \rfloor \geq \lfloor \log_p(k/2) \rfloor$, this improves the $k/2$ bound from the discussion thread (as does rickyc's $k - 1$ version there), with strict improvement when $p^{L_p} > k/2$ and $k \neq p^{L_p}$: for $k = 6, 14, 18, 54$, $W'_k = 72, 784, 2592, 46656$ against $36, 392, 1296, 23328$. (ii) For prime k the bound is only $n_k \geq k$. It is not comparable with the claimed $c(\log k)^2$ of (1) without an explicit c; the data force $c \leq 0.740$ (from $k = 19$), and with that value $\log W'_k$ exceeds $c(\log k)^2$ for 28 of the 80 composite $k \leq 110$, those with several small prime factors. Its merit is that it is explicit and attained, not that it is large. (iii) Along primorials $k = p_1 \cdots p_m$, $W'_k > k^{\omega(k)} k / \mathrm{rad}(k)$ gives $n_k \geq \exp\big((1 + o(1))(\log k)^2 / \log\log k\big)$.

4. Primes larger than $\sqrt{k}$

Lemma 4.1. *Let p be a prime with $\sqrt{k} < p \leq k$, so $L_p = 1$ and $p^2 > k$, and let $j = \lfloor k/p \rfloor$.*

1. *If $n \bmod p < k \bmod p$, then $\mathrm{Bad}_p(n) = \varnothing$.*
2. *Otherwise, if $j = 1$, the block contains exactly one multiple of p, at index $n \bmod p$, and $\mathrm{Bad}_p(n) = \{n \bmod p\}$.*
3. *Otherwise ($j \geq 2$), the block contains exactly j multiples of p, at most one of which is divisible by p^2; $|\mathrm{Bad}_p(n)| = 1$ iff one of them is divisible by p^2, and then that index is the bad one; if none is, all j are bad.*

Consequently (3) *holds.*

Proof. With $L_p = 1$, $d_p = [\,n \bmod p \geq k \bmod p\,]$. If $n \bmod p < k \bmod p$ then $d_p = 0$ and $\mathrm{Bad}_p = \varnothing$ by Lemma 2.3(1). Otherwise $d_p = 1$, so $\mathrm{Bad}_p = \{i : v_p(n-i) \geq M_p\}$ is the set of indices of maximal valuation, and by Lemma 2.2 the block contains exactly $j + \delta_1 = j$ multiples of p. If $j = 1$, the unique multiple of p has index $i \equiv n \pmod p$ with $0 \leq i < p \leq k$, i.e. $i = n \bmod p$; its valuation is $M_p \geq 1$ and all other valuations are 0. If $j \geq 2$, two multiples of p^2 in the block would differ by at least $p^2 > k$, so at most one index has valuation ≥ 2; if there is one it is the unique maximum, and otherwise all j multiples of p tie at valuation $1 = M_p$.

For (3): the events $d_p = 0$ and $d_p = 1$ are disjoint, and $\Pr(d_p = 0) = (k \bmod p)/p$. Given $d_p = 1$: if $j = 1$, $\mathrm{Bad}_p = \{i_0\}$ iff $n \bmod p = i_0$, which requires $i_0 < p$ and $i_0 \geq k \bmod p$ and has probability $1/p$ (if $i_0 \geq p$ then $i_0 \bmod p = i_0 - p < k - p = k \bmod p$, so the indicator in (3) vanishes, as it should); if $j \geq 2$, $\mathrm{Bad}_p = \{i_0\}$ iff $p^2 \mid n - i_0$, which has probability $1/p^2$ and implies $n \bmod p = i_0 \bmod p$, so that $d_p = 1$ is exactly the condition $i_0 \bmod p \geq k \bmod p$. □

5. Periodicity and the exact density

Proof of Theorem 1.3. Step 1: $\mathrm{Bad}_p(n)$ *depends only on* $n \bmod p^{2L}$, *where* $L = L_p$. Let $n' \equiv n \pmod{p^{2L}}$. The deficiency d_p depends only on $n \bmod p^L$, so $d_p(n') = d_p(n)$. For each index i, $n' - i \equiv n - i \pmod{p^{2L}}$, so $v_p(n'-i) = v_p(n-i)$ whenever $v_p(n-i) < 2L$, and $v_p(n'-i) \geq 2L$ iff $v_p(n-i) \geq 2L$. Two indices with valuation $\geq L+1$ would differ by a multiple of $p^{L+1} > k$, so at most one index has valuation $\geq L+1$. If no index has valuation $\geq 2L$, all valuations agree between n and n', and so do M_p, d_p and Bad_p. If one index i^* has $v_p(n-i^*) \geq 2L$, it is the unique maximum for both n and n', and every other valuation is $\leq L$ and agrees; by Lemma 2.3, i^* is bad iff $d_p \geq 1$, for n and n' alike, while another index i would need $v_p(n-i) > M_p - d_p \geq 2L - L = L$, which is impossible. So $\mathrm{Bad}_p(n') = \mathrm{Bad}_p(n)$, and E_{p,i_0} is a union of residue classes modulo p^{2L}.

The modulus p^{2L-1} does not suffice in general. For $(k,p) = (6,3)$: the block of $n = 12$ is $\{7,\dots,12\}$ with $v_3(9) = 2$ at $i = 3$ and $v_3(12) = 1$ at $i = 0$, so $M_3 = 2$, $d_3 = [12 \bmod 3 \geq 6 \bmod 3] = 1$ and $\mathrm{Bad}_3 = \{3\}$; the block of $n = 15 \equiv 12 \pmod 3$ is $\{10,\dots,15\}$ with $v_3(12) = v_3(15) = 1$, so $M_3 = 1$, $d_3 = 1$ and $\mathrm{Bad}_3 = \{0,3\}$.

Step 2: admissibility. By Lemma 2.1 the set of bad indices is $\bigcup_{p\leq k} \mathrm{Bad}_p(n)$, and it is nonempty by Corollary 2.4. So n is admissible iff there is an i_0 with $\mathrm{Bad}_p(n) \subseteq \{i_0\}$ for every $p \leq k$, i.e. $n \in \bigcap_{p\leq k} E_{p,i_0}$. If this held for two indices $i_0 \neq i_1$, every $\mathrm{Bad}_p(n)$ would be empty, contradicting Corollary 2.4; so the k sets $\bigcap_p E_{p,i_0}$ are pairwise disjoint.

Step 3: density. By Step 1 and the Chinese remainder theorem, $\bigcap_{p\leq k} E_{p,i_0}$ is a union of residue classes modulo D_k, and the proportion of classes it contains is $\prod_{p\leq k} \Pr(E_{p,i_0})$, since $n \bmod D_k \mapsto (n \bmod p^{2L_p})_{p\leq k}$ is a bijection. The natural density of a periodic set is its proportion of classes per period, and by Step 2 the admissible n form the disjoint union over i_0; the restriction $n \geq 2k$ does not affect density. The closed form (3) is Lemma 4.1. □

For $p \leq \sqrt{k}$ the factor $\Pr(E_{p,i_0})$ can be computed by enumerating $n \bmod p^{2L_p}$; our implementation enumerates $n \bmod p^{L_p+1}$ and treats the valuation of the at most one multiple of p^{L_p+1} as a geometric tail. Two independent implementations agree, and brute-force counts over full periods confirm, for instance, $\delta(7) = 269/2520$, $\delta(11) = 9803/831600$ and $\delta(12) = 1/1386$.

6. The size of the density

Throughout this section p denotes a prime, $\{x\} = x - \lfloor x \rfloor$, and for a prime $p \in (\sqrt{k}, k]$ with $p \nmid k$ we write

$$j = j(p) = \lfloor k/p \rfloor, \qquad r = r(p) = k - jp = k \bmod p \in [1, p-1],$$

so that $\{k/p\} = r/p$ and $1 \le j < \sqrt{k}$. The at most one prime $p > \sqrt{k}$ dividing k is excluded from $S(k)$ because $k \bmod p = 0$; Part 1 below shows it contributes $O(\log k)$ to $\log(1/\delta(k))$. We use three classical inputs: the prime number theorem in the form $\pi(x) = \mathrm{li}(x) + E(x)$ with $E(x) \ll x\exp(-c\sqrt{\log x})$ [4]; the Brun–Titchmarsh inequality in the form of Montgomery and Vaughan [11], $\pi(x+y) - \pi(x) \le 2y/\log y$ for integers $x \ge 0$, $y \ge 2$, which for a real interval of length $y \ge 2$ (contained in an integer interval of length at most $y+1$) gives

$$\pi(x+y) - \pi(x) \le \frac{3y}{\log y} \qquad (x \ge 0,\ y \ge 2 \text{ real}), \tag{4}$$

since $2(y+1)/\log(y+1) \le 3y/\log y$ for $y \ge 2$; and Chebyshev's bounds $\pi(x) \ll x/\log x$, $\vartheta(x) \ll x$.

Lemma 6.1 (injectivity)**.** *The map $p \mapsto k \bmod p$ is injective on the primes $p \in (\sqrt{k}, k]$ with $p \nmid k$.*

Proof. If $p \ne q$ are two such primes with $k \bmod p = k \bmod q = r$, then $pq \mid k - r$; but $k - r = jp \ge p > 0$ and $k - r \le k < pq$. □

Lemma 6.2 (the constant)**.** *For $j \ge 1$, $c_j := \int_j^{j+1} \frac{-\log(x-j)}{x^2}\,dx = \frac{1}{j}\log\left(1+\frac{1}{j}\right)$, and $\sum_{j\ge1} c_j = C$ with $\sum_{j>J} c_j \le 1/J$.*

Proof. Substituting $u = x - j$, $c_j = \int_0^1 \frac{-\log u}{(j+u)^2}\,du$. Integrate by parts with $v(u) = \frac{u}{j(j+u)}$, which satisfies $v'(u) = (j+u)^{-2}$ and $v(0) = 0$:

$$c_j = \Big[-\log u \cdot \frac{u}{j(j+u)}\Big]_0^1 + \int_0^1 \frac{1}{u}\cdot\frac{u}{j(j+u)}\,du = \frac{1}{j}\int_0^1 \frac{du}{j+u} = \frac{1}{j}\log\frac{j+1}{j},$$

the boundary term vanishing since $\log 1 = 0$ and $u\log u \to 0$. As $\log(1+1/j) \le 1/j$, $c_j \le j^{-2}$ and $\sum_{j>J} c_j \le 1/J$; splitting $[1,\infty)$ at the integers gives $\sum_j c_j = \int_1^\infty -\log\{x\}\,x^{-2}dx$. The expansion $\log(1+1/j) = \sum_{m\ge1}(-1)^{m+1}j^{-m}/m$ gives the zeta form of C: the $j=1$ term is $\log 2 = \sum_m (-1)^{m+1}/m$ (conditionally convergent), and for $j \ge 2$ the double series converges absolutely, since $\zeta(m+1) - 1 \le 3\cdot 2^{-m-1}$, so the sums may be interchanged. □

Part 1: $\log(1/\delta(k)) = S(k) + O(\sqrt{k})$**.** *Upper bound for δ.* In (2), bound each of the k summands by the product over the primes $p > \sqrt{k}$, $p \nmid k$ only (the other factors are at most 1), and each such factor, using (3), by $\{k/p\} + 1/p$:

$$\delta(k) \le k \prod_{\substack{\sqrt{k}<p\le k\\ p\nmid k}} \Big(\Big\{\frac{k}{p}\Big\} + \frac{1}{p}\Big) = k\prod\Big\{\frac{k}{p}\Big\}\prod\Big(1 + \frac{1}{k \bmod p}\Big).$$

By Lemma 6.1 the values $k \bmod p$ are distinct positive integers at most k, so $\sum \log(1+\frac{1}{k \bmod p}) \le \sum_{r\le k}\frac{1}{r} \le \log k + 1$. Hence $\log(1/\delta(k)) \ge S(k) - 2\log k - 1$.

Lower bound for δ. Keep only the term $i_0 = 0$ in (2). For $p \le \sqrt{k}$ (so $L = L_p \ge 1$): if $p^{2L} \mid n$ then $v_p(n) \ge 2L$ while $v_p(n-i) = v_p(i) \le L < 2L$ for $1 \le i < k$ (as $i < k < p^{L+1}$); so $M_p = v_p(n)$ is attained only at $i = 0$, and every other index has valuation $\le L \le M_p - d_p$ because $d_p \le L$; by Lemma 2.3, $\mathrm{Bad}_p(n) \subseteq \{0\}$. Thus $\Pr(E_{p,0}) \ge p^{-2L} \ge k^{-2}$, and $\sum_{p\le\sqrt{k}} 2\log k \ll \sqrt{k}$ by Chebyshev. For $\sqrt{k} < p \le k$ with $p \nmid k$, (3) gives $\Pr(E_{p,0}) \ge \{k/p\}$. For the at most one prime $p > \sqrt{k}$ with $p \mid k$, $k \bmod p = 0$ and (3) gives $\Pr(E_{p,0}) \ge 1/p^2 \ge 1/k^2$. Hence $\log(1/\delta(k)) \le S(k) + O(\sqrt{k})$.

Part 2: $S(k) = (C + o(1))\, k/\log k$. Fix $\varepsilon \in (0, 1/e)$ and an integer $J \geq 2$; they stay fixed while $k \to \infty$ (assume $k > (J+1)^2$), and are sent to 0 and ∞ at the end. Cover the primes $p \in (\sqrt{k}, k]$, $p \nmid k$, by three classes according to $r = k \bmod p$ and $j = \lfloor k/p \rfloor$:

$$\mathcal{T}:\ r \leq 4j, \qquad \mathcal{D}:\ 4j < r \leq \varepsilon p, \qquad \mathcal{B}:\ r > \varepsilon p.$$

($\mathcal{T}$ and $\mathcal{B}$ can overlap for $p < 2\sqrt{k/\varepsilon}$. This is harmless: the upper bound on $S(k)$ only needs a cover, and the lower bound uses only the part of $\mathcal{B}$ with $j \leq J$, which is a subset of the terms.)

Class $\mathcal{T}$. For primes with $r \leq 4j$ we have $jp \leq k-1$ (as $r \geq 1$) and $k \leq jp + 4j$, so $p \in [k/j - 4,\ k/j)$, a half-open interval of length 4 containing at most 4 integers. So there are at most 4 such primes for each of the fewer than $\sqrt{k}$ values of j, and every term is at most $\log p \leq \log k$. The class contributes at most $4\sqrt{k}\log k = o(k/\log k)$.

Class $\mathcal{D}$. Group the primes by j and by dyadic ranges $r \in (R, 2R]$ with R a power of 2. A prime in the group satisfies $k - 2R \leq jp < k - R$, so p lies in an interval of length R/j; and $r > 4j$ forces $R > 2j$, so $R/j > 2$ and (4) applies: the group has at most $\frac{3R/j}{\log(R/j)}$ primes. Each term is $\log(p/r) \leq \log \frac{k}{jR}$, since $p \leq k/j$ and $r > R$. Also $R < r \leq \varepsilon p \leq \varepsilon k/j$.

Put $V = \varepsilon k/j^2$ and, for a group with range $(R, 2R]$, $u = R/j$; then $2 < u < V$ and $\log \frac{k}{jR} = \log \frac{V}{u} + \log \frac{1}{\varepsilon}$. So the contribution of fixed j is at most

$$\sum_u \frac{3u}{\log u}\Big(\log \frac{V}{u} + \log \frac{1}{\varepsilon}\Big),$$

where u runs over the values R/j with R a power of two and $2 < u < V$. The admissible R are the powers of two below $\varepsilon k/j$; the largest, $R_{\max}$, satisfies $R_{\max} \geq \varepsilon k/(2j)$, so the u are $V'/2^m$ ($m \geq 0$) with $V/2 \leq V' < V$, and $V/2^{m+1} \leq u \leq V/2^m$.

- For $m \leq \frac{1}{2}\log_2 V - 1$: here $\log u \geq \log(V/2^{m+1}) \geq \frac{1}{2}\log V$ and $\log(V/u) \leq (m+1)\log 2$, so these terms sum to at most $\frac{6V}{\log V}\sum_{m\geq 0} \frac{(m+1)\log 2 + \log(1/\varepsilon)}{2^m} \leq \frac{24V}{\log V}\big(\log 2 + \log \frac{1}{\varepsilon}\big)$.
- For $m > \frac{1}{2}\log_2 V - 1$: here $u \leq V/2^m < 2\sqrt{V}$, while $u > 2$ gives $\log u \geq \log 2$ and $\log(V/u) \leq \log V$. Each such group contributes at most $\frac{6\sqrt{V}}{\log 2}\big(\log V + \log \frac{1}{\varepsilon}\big)$, and there are at most $\log_2 V$ such groups (as $u > 2$ forces $2^m < V/2$). So these terms sum to $\ll \sqrt{V}(\log k)^2$, as $V \leq k$ and ε is fixed.

Now sum over j; only j with $V > 2$, i.e. $j < \sqrt{\varepsilon k/2}$, contribute. The $\sqrt{V}$ terms give $\ll \sqrt{\varepsilon k}(\log k)^2 \sum_{j<\sqrt{k}} \frac{1}{j} \ll \sqrt{k}(\log k)^3 = o(k/\log k)$. For $j \leq k^{1/4}$, $\log V \geq \log(\varepsilon\sqrt{k}) \geq \frac{1}{4}\log k$ for large k, so $\sum_{j\leq k^{1/4}} \frac{24V}{\log V}(\log 2 + \log \frac{1}{\varepsilon}) \leq \frac{96\varepsilon k}{\log k}(\log 2 + \log \frac{1}{\varepsilon}) \sum_j j^{-2} \ll \varepsilon \log(1/\varepsilon)\frac{k}{\log k}$. For $k^{1/4} < j < \sqrt{\varepsilon k/2}$, $\log V \geq \log 2$, so these give $\ll \varepsilon k \log(1/\varepsilon) \sum_{j>k^{1/4}} j^{-2} \ll \varepsilon \log(1/\varepsilon) k^{3/4} = o(k/\log k)$. Altogether, class $\mathcal{D}$ contributes at most $A\,\varepsilon \log(1/\varepsilon)\, k/\log k + o(k/\log k)$ for an absolute constant A.

Class $\mathcal{B}$, $j > J$. Each term is $\log(p/r) < \log(1/\varepsilon)$. The primes with $j > J$ satisfy $p < k/J$, so by (4) with $x = 0$ there are at most $\pi(k/J) \leq \frac{3k}{J\log(k/J)} \leq \frac{6k}{J\log k}$ of them for $k \geq J^2$. The contribution is at most $6\log(1/\varepsilon)\, k/(J\log k)$.

Class $\mathcal{B}$, $j \leq J$: the main term. Fix $j \leq J$. The primes of class $\mathcal{B}$ with $\lfloor k/p \rfloor = j$ are the primes in the interval

$$I_j(\varepsilon) = \Big(\frac{k}{j+1},\ \frac{k}{j+\varepsilon}\Big),$$

because $r > \varepsilon p \iff k > (j+\varepsilon)p$. (A prime $p \mid k$ with $\lfloor k/p \rfloor = j$ equals k/j, which is not in $I_j(\varepsilon)$, so the condition $p \nmid k$ excludes nothing here; and $I_j(\varepsilon) \subset (\sqrt{k}, k]$ since $k > (J+1)^2$.) On

this interval put $g(t) = \log \frac{t}{k-jt} = -\log\big(\frac{k}{t} - j\big)$; then $0 < g(t) < \log(1/\varepsilon)$ and $g'(t) = \frac{1}{t} + \frac{j}{k-jt} \le \frac{1+j/\varepsilon}{t} \le \frac{(J+1)(1+J/\varepsilon)}{k}$.

By partial summation with $\pi(t) = \mathrm{li}(t) + E(t)$, writing $a < b$ for the endpoints of $I_j(\varepsilon)$,

$$\sum_{p \in I_j(\varepsilon)} g(p) = \int_a^b \frac{g(t)}{\log t}\,dt + \big[g(t)E(t)\big]_a^b - \int_a^b g'(t)E(t)\,dt + O\big(\log\tfrac{1}{\varepsilon}\big),$$

where the $O(\cdot)$ covers the endpoints. The two error terms are at most $2\log(1/\varepsilon)\max_{[a,b]}|E| + (b-a)\max_{[a,b]}|g'|\max_{[a,b]}|E| \ll_{\varepsilon,J} k\exp(-c'\sqrt{\log k}) = o(k/\log k)$, since $[a,b] \subset [k/(J+1), k]$. For the main integral, $\log t = \log k + O(\log(J+1)) = (1+o(1))\log k$ on $[a,b]$, so, substituting $x = k/t$, $dt = -k\,x^{-2}dx$,

$$\int_a^b \frac{g(t)}{\log t}\,dt = \frac{(1+o(1))\,k}{\log k}\int_{j+\varepsilon}^{j+1} \frac{-\log(x-j)}{x^2}\,dx =: \frac{(1+o(1))\,k}{\log k}\,c_j(\varepsilon).$$

Here $0 \le c_j - c_j(\varepsilon) = \int_j^{j+\varepsilon} \frac{-\log(x-j)}{x^2}dx \le \frac{1}{j^2}\int_0^{\varepsilon} -\log u\,du = \frac{\varepsilon(1+\log(1/\varepsilon))}{j^2} \le \frac{2\varepsilon\log(1/\varepsilon)}{j^2}$ for $\varepsilon < 1/e$. Summing over $j \le J$, with Lemma 6.2 and $\sum_j j^{-2} < 2$,

$$\sum_{\substack{p \in \mathcal{B} \\ j(p) \le J}} \log\frac{p}{r} = \frac{(1+o(1))\,k}{\log k}\sum_{j\le J} c_j(\varepsilon), \qquad C - \frac{1}{J} - 4\varepsilon\log\frac{1}{\varepsilon} \;\le\; \sum_{j\le J} c_j(\varepsilon) \;\le\; C.$$

Conclusion. All terms of $S(k)$ are positive, so the $j \le J$ part of class $\mathcal{B}$ alone gives the lower bound, and the four pieces together give the upper bound:

$$\Big(C - \frac{1}{J} - 4\varepsilon\log\frac{1}{\varepsilon} - o(1)\Big)\frac{k}{\log k} \;\le\; S(k) \;\le\; \Big(C + A\varepsilon\log\frac{1}{\varepsilon} + \frac{6\log(1/\varepsilon)}{J} + o(1)\Big)\frac{k}{\log k},$$

where $o(1) \to 0$ as $k \to \infty$ with ε, J fixed. Letting $k \to \infty$, then $J \to \infty$, then $\varepsilon \to 0$, gives $S(k) = (C + o(1))\,k/\log k$. With Part 1 this proves Theorem 1.4. □

Remark 6.3. The convergence is slow. Averaged over 21 consecutive values of k, $S(k)/(k/\log k)$ is 1.16 near $k = 100$, 1.37 near 10^4, and 1.34 near 10^6; the next term should be $c_2 k/\log^2 k$ with $c_2 = \int_1^\infty -\log\{x\}\log x\,x^{-2}dx = 1.0016551589\ldots$, obtainable by keeping the variation of $\log t$ in the main integral. The class bounds are visible numerically: the class $\mathcal{D}$ sum divided by $\varepsilon\log(1/\varepsilon)k/\log k$ is $2.29, 2.08, 2.10, 1.97$ at $(k,\varepsilon) = (10^6, 0.1), (10^6, 0.02), (10^7, 0.02), (10^7, 0.005)$, so $A \approx 2$ in practice. The small-prime part $\log(1/\delta(k)) - S(k)$ lies between 3.2 and 17.0 for $31 \le k \le 110$.

7. The growth conjecture and the evidence

Theorem 1.4 concerns the density of admissible n. If the admissible set behaved like a random set of density $\delta(k)$, its least element would be about $\mathrm{Exp}(1)/\delta(k)$; this is Conjecture 1.5. It is not a theorem, and we do not know how to prove either inequality in it (see Section 8). The evidence is as follows.

New values. Table 2 gives n_k for $60 \le k \le 110$, extending the OEIS table. The values for $60 \le k \le 75$ were posted earlier in the discussion thread by jbell31 (28 July 2026), from a search that assumes rickyc's divisibility result (now formally verified; see the end of Section 3); ours were obtained independently and agree with all 16. Each value was obtained by an exhaustive search over the candidate set defined by the necessary conditions of Proposition 3.1 and Lemma 4.1, with every candidate tested by the exact criterion of Lemmas 2.1–2.2; the same program reproduces the 58 OEIS values for $k \le 59$. Every value was then verified with exact

integer arithmetic (exactly one of $n, \dots, n-k+1$ fails to divide $\binom{n}{k}$), and minimality was confirmed by a second, independently written exhaustive search, and for several k by a plain scan from $2k$.

Prime versus composite k. The model reproduces the large swings between neighbouring k: primes dividing k force $n - i_0$ to be divisible by $p^{a+\lfloor \log_p(k-1) \rfloor}$ (Proposition 3.1), which is why $n_{30} \approx 3.8 \cdot 10^7$ while $n_{31} = 8100$.

Fit. Let $x_k = \log n_k - \log(1/\delta(k))$. For log of an Exp(1) variable the mean is $-\gamma = -0.577$ and the standard deviation $\pi/\sqrt{6} = 1.283$. For $31 \le k \le 110$ (80 values; sample standard deviations throughout) x_k has mean -0.58 and standard deviation 1.19, a Kolmogorov–Smirnov statistic $D = 0.083$ against $\log \mathrm{Exp}(1)$ (5% critical value 0.152), and $\log n_k$ and $\log(1/\delta(k))$ have correlation 0.988 with regression slope 0.985. For $k \le 30$ the fit is poor (mean $+0.19$, $D = 0.348$); there the constraint $n \ge 2k$ and the coarse structure of the admissible set matter.

Out-of-sample tests. Before computing any n_k with $k \ge 60$, the central 90% interval of $\mathrm{Exp}(1)/\delta(k)$, namely $[0.0513/\delta(k),\ 2.996/\delta(k)]$, was recorded for every $60 \le k \le 100$; 38 of the 41 values fell inside (outside: $k = 64, 91, 95$). The same was then done for $101 \le k \le 110$, with 9 of 10 inside (outside: $k = 105$). Combined: 47 of 51, against 45.9 expected. The model has no fitted parameters, so the order of events could not have influenced the predictions; the timestamped prediction files are in [3].

Beyond the least element. For $k \in \{40, 45, 50, 61, 64, 67\}$ we enumerated the first 139–168 admissible n (Table 1). The m-th admissible number is close to $m/\delta(k)$ (median of $m\,\delta(k)\,n^{(m)}$ between 0.90 and 1.22), the count below X is within 11% of $\delta(k)X$, the gaps between consecutive failing elements $n - i_0$ have coefficient of variation 0.84–1.02 (exponential: 1), and the observed frequencies of the failing index i_0 match the model's predicted weights $\prod_p \Pr(E_{p,i_0})$. So the model describes the whole admissible set, not only its least element; the least element is the noisy statistic (for $k = 64$ it is 20 times below $1/\delta$, the outlier $x_{64} = -3.0$ above). For $k \le 30$ the admissible n cluster (e.g. $n_{30} = 37584001$ and 37584002 share the failing element 37584000), which is consistent with the worse fit at small k; for $k \ge 40$ clusters are rare. Finally, an admissible n lands in the carry-free window $n \bmod p < k \bmod p$ at all but on average 0.4–0.7 of the primes $p \in (\sqrt{k}, k]$, $p \nmid k$, the exceptions being almost always primes $p > k/2$ dividing $n - i_0$. This is the structure a construction would have to reproduce (Section 8).

TABLE 1. The first admissible n beyond n_k: N values enumerated, the median of $m\,\delta(k)\,n^{(m)}$ over $m \le N$ (model: ≈ 1), the ratio of the count below $X = n^{(N)}$ to $\delta(k)X$, the coefficient of variation of the gaps between consecutive failing elements, and the mean number of large primes ($\sqrt{k} < p \le k$, $p \nmid k$) at which n is *not* in the carry-free window.

k	N	median $m\delta n^{(m)}$	count/δX	gap CV	non-window primes per n
40	139	1.05	0.95	0.97	0.43
45	139	1.22	0.94	0.90	0.67
50	150	1.08	0.98	0.94	0.48
61	168	0.90	1.11	0.94	0.66
64	150	0.99	1.00	1.02	0.54
67	161	0.98	1.06	0.84	0.41

8. Relation to the claimed bounds of Cipollini

If both claims in (1) hold, then Conjecture 1.5 sits strictly between them and says that the upper bound is off by a factor of order $\log\log k$. In the data, for $31 \le k \le 100$, $\log n_k$ is between 0.47 and 0.89 (mean 0.68) of the main term $\frac{k}{\log k}(\log\log k + \log\log\log k + \log 2)$.

The local lemmas of [1] (its Lemma 2, "stabilization", and Lemma 3, "one carry") are the same facts as our Lemma 2.3 and Lemma 4.1; Lemma 3 of [2], that $k \mid n - i_0$ for admissible n, is a special case of Proposition 3.1. The construction in [1] takes $n = e + Mt$ with a small exceptional index $e \approx k/(2\log k \log\log k)$, forces the primes dividing $k\binom{k-1}{e}$ by divisibility, and uses a pigeonhole (simultaneous approximation) to put $n \bmod p$ into the carry-free window $[0, k \bmod p)$ for the remaining large primes. Pigeonhole on differences yields symmetric windows, of half-width $\min(e, (k \bmod p) - e) \le e$, at a cost of $\log(p/e)$ per prime; summed over primes this is $\frac{k}{\log k}(\log(k/e) - 1 + o(1))$. The density model's natural window is the full $[0, k \bmod p)$, at cost $\log \frac{p}{k \bmod p}$ per prime, and these costs sum to $S(k) \sim Ck/\log k$. The enumeration in Section 7 shows that actual admissible n do sit in the one-sided window at essentially every large prime. The gap between the two costs is exactly the gap between what a symmetric pigeonhole can reach and what a one-sided target would require; closing it appears to need an equidistribution-type input for the vectors $(n \bmod p)_p$ over $n \le e^{(C+\varepsilon)k/\log k}$, a range far smaller than the product of the moduli, and we do not know how to obtain one. The lower-bound half of the conjecture, that no admissible n lies below $e^{(C-\varepsilon)k/\log k}$, is a least-element problem of Linnik type and seems out of reach.

Appendix A. The values n_k for $60 \le k \le 110$

Table 2. n_k and the bad index i_0 (so $n_k - i_0$ is the element of the block that fails to divide $\binom{n_k}{k}$).

k	n_k	i_0	k	n_k	i_0	k	n_k	i_0
60	2 117 441 088 029	29	61	569 181 900	50	62	155 275 274 056	8
63	216 949 362 182	35	64	2 600 001 558	22	65	1 936 512 990 784	34
66	1 618 958 347 825	49	67	1 178 168 706	37	68	260 686 545 186	34
69	709 479 687 556	28	70	66 383 308 992 066	66	71	13 760 436 550	35
72	890 473 430 071	55	73	40 861 892 736	21	74	3 146 202 243 081	9
75	32 639 932 074 432	57	76	16 233 110 681 625	25	77	18 874 715 290 176	53
78	780 422 752 841 501	29	79	420 957 410 952	14	80	60 137 833 600 000	0
81	2 418 072 694 480	46	82	94 878 375 215 625	9	83	885 437 574 400	11
84	3 765 171 612 545 280	0	85	1 718 626 725 667 152	27	86	129 713 628 198 225	81
87	2 221 164 774 687 826	82	88	754 618 137 360 426	42	89	3 208 190 120 080	80
90	57 753 167 368 032 004	4	91	10 789 171 853 293 890	56	92	2 643 061 175 354 691	67
93	7 201 679 510 714 780	29	94	2 544 131 371 430 937	25	95	213 706 678 047 814	64
96	11 795 043 698 546 775	87	97	10 432 185 034 560	35	98	23 154 863 741 528 961	1
99	7 525 993 870 612 546	43	100	24 275 936 775 680 073	73	101	83 339 504 740 900	4
102	455 022 751 908 409 400	56	103	80 855 126 198 850	57	104	28 292 410 857 118 753	33
105	114 184 320 231 814 976	101	106	505 139 783 269 972 305	81	107	314 604 394 073 026	96
108	42 039 517 780 512 072	72	109	92 095 914 818 376	95	110	4 499 581 121 078 560 101	101

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